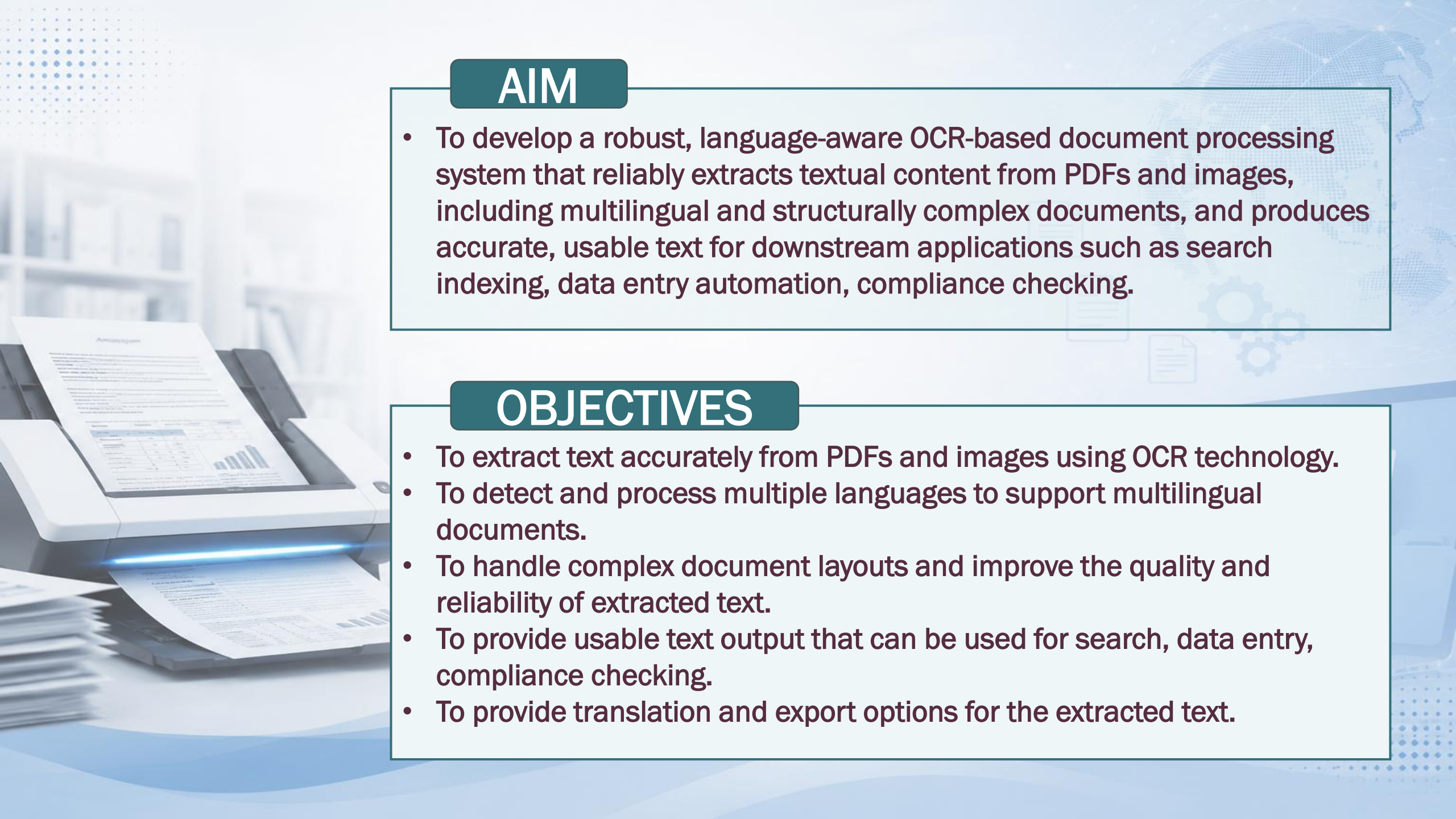


Multilingual Document OCR Extraction Pipeline

**Presented By - Bhakti Patel
Shrusti Dalwadi**

Problem statement

- Documents need to have their textual content extracted reliably so that the extracted text can be consumed by downstream systems (search indexing, data entry automation, compliance checks, analytics).
- The absence of a robust, language-aware OCR pipeline means multilingual and structurally complex documents either fail to process or produce low-quality, unreliable text output.

The background of the slide features a document scanner on the left side, with a document being processed. The document has a header and a bar chart. On the right side, there is a faint, stylized globe with gears, symbolizing technology and automation. The overall color scheme is light blue and white.

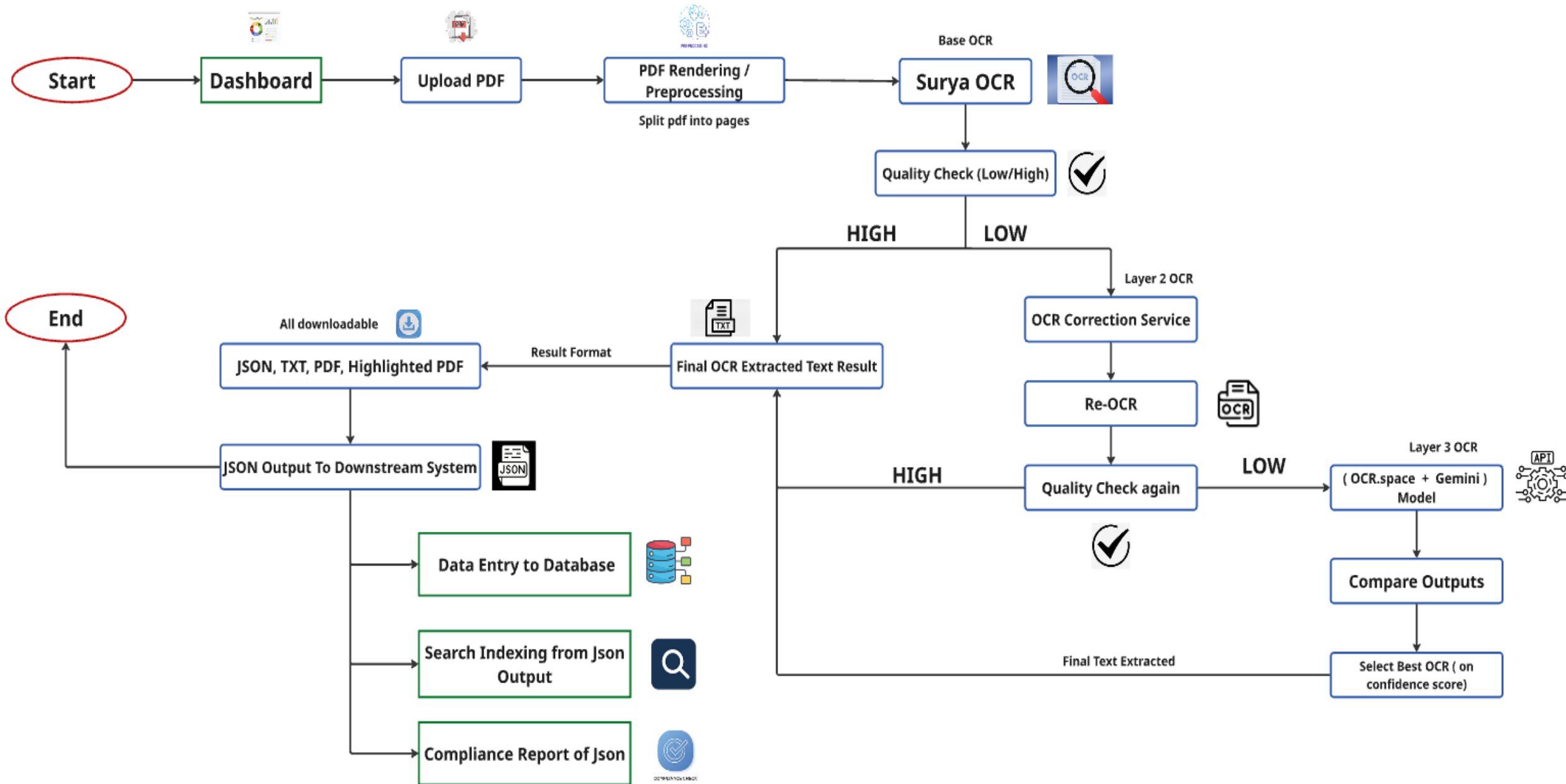
AIM

- To develop a robust, language-aware OCR-based document processing system that reliably extracts textual content from PDFs and images, including multilingual and structurally complex documents, and produces accurate, usable text for downstream applications such as search indexing, data entry automation, compliance checking.

OBJECTIVES

- To extract text accurately from PDFs and images using OCR technology.
- To detect and process multiple languages to support multilingual documents.
- To handle complex document layouts and improve the quality and reliability of extracted text.
- To provide usable text output that can be used for search, data entry, compliance checking.
- To provide translation and export options for the extracted text.

Work Flow Diagram



Advantages

1. **Saves Time & Effort** – Automatically extracts information from multi-page PDFs and images, reducing repetitive manual typing and document processing time.
2. **Multilingual Processing** – Processes multilingual documents including English, Hindi and Gujarati, enabling efficient extraction across different languages.
3. **Structure-Aware Extraction** – Preserves document layout, bounding boxes and reading order, providing more meaningful and organized information than plain text.
4. **Multiple Structured Outputs** – Generates JSON, TXT and searchable PDF from the same document, making the extracted data useful for both users and applications.
5. **Downstream Automation** – Extracted information can directly support search, data entry, compliance checking and analytics, extending OCR beyond simple text extraction.
6. **Open-Source & Customizable** – Uses open-source components with control over the processing pipeline, allowing customization and deployment without proprietary OCR platform dependency.

Challenges

1. **Multilingual OCR Accuracy** – Handling different scripts such as English, Hindi and Gujarati while maintaining accurate text recognition. Mixed-language documents and varied fonts, image quality and characters made extraction more challenging.
2. **Complex Document Layouts** – Extracting text while preserving reading order, tables, positions and document structure. Simple OCR output was not sufficient for documents containing multiple sections, columns and structured content.
3. **Choosing the Right OCR Architecture** – Evaluating Tesseract, Indic-focused approaches, PaddleOCR and Surya for our requirements. Finding an approach that balanced multilingual support, layout analysis and open-source flexibility required experimentation.
4. **Converting OCR into Usable Data** – Designing a pipeline that transforms raw OCR results into reliable JSON, TXT and searchable PDF outputs. The structured output also had to be suitable for downstream **search, data entry, compliance and analytics**.

What Makes Our Solution Different

Existing	Our solution
Platform / service dependent	Open-source, customizable OCR pipeline
Structured outputs with platform-based workflows	Structured JSON, TXT & PDF built for downstream systems
Processing pipeline is constrained by the platform	Our pipeline controls preprocessing, layout analysis & page splitting
Translation may require an additional service/tool	Integrated translation after text extraction

RESEARCH POINTS

- **Document OCR is more than text recognition.** Reliable document processing involves OCR along with layout analysis, reading order, tables, and document structure.
- **Multilingual and complex documents require more than conventional OCR.**
- **OCR output alone is not sufficient for downstream systems;** extracted information needs to become **structured, searchable, and application-ready.**
- **Evaluated multiple open-source OCR approaches** including Tesseract, Indic-focused solutions, and PaddleOCR to understand multilingual recognition, preprocessing, and document-processing capabilities.
- **Selected Surya** because its OCR, layout analysis, reading-order detection, and multilingual capabilities aligned closely with our document-processing requirements.
- **Built an end-to-end pipeline around Surya** that transforms multi-page documents into structured, machine-readable outputs for downstream systems.

REFERENCES

- [2206.01062] DocLayNet: A Large Human-Annotated Dataset for Document-Layout Analysis
- datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages
- [2510.14528] PaddleOCR-VL: Boosting Multilingual Document Parsing via a 0.9B Ultra-Compact Vision-Language Model
- Text extraction using ocr:- <https://ieeexplore.ieee.org/abstract/document/9183326>
- A Survey of Deep Learning Approaches for OCR and Document Understanding:- <https://arxiv.org/abs/2011.13534>

Thank you 😊

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